Hints: Methods of proof – 2

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Lines: the n-th line meets the previous $n-1$ lines in $n-1$ distinct points. These points cut it into n pieces, and each piece splits one old region into two. So $f(n) = f(n-1) + n$ with $f(0) = 1$, which gives $f(n) = 1 + \frac{n(n+1)}{2} = \frac{n^2+n+2}{2}$.

Hint 2. In $\mathbb{R}^d$, let $f(n, d)$ be the number of regions cut out by n hyperplanes in general position. The n-th hyperplane $H \cong \mathbb{R}^{d-1}$ is cut by the others into $f(n-1, d-1)$ regions, because their traces on H are $n-1$ hyperplanes of H in general position. Each such region splits one old region.

Hint 3. So $f(n, d) = f(n-1, d) + f(n-1, d-1)$, with $f(0, d) = 1$ and $f(n, 0) = 1$. This is Pascal's rule, and the double induction on (n, d) (e.g. on $n + d$) gives $f(n, d) = \sum_{i=0}^{d} \binom{n}{i}$.

Problem 2

Hint 1. Let $\Delta P(x) = P(x+1) - P(x)$. If $\deg P = n$ and P is integer at $a, a+1, \ldots, a+n$, then ΔP has degree $n-1$ and is integer at $a, \ldots, a+n-1$. Induct on the degree: all $\Delta^k P(a)$, $k = 0, \ldots, n$, are integers, and $\Delta^{n+1} P = 0$.

Hint 2. Newton's forward-difference formula: $P(x) = \sum_{k=0}^{n} \Delta^k P(a) \binom{x-a}{k}$. Prove it by induction on n too, applying it to ΔP and using $\Delta\binom{x-a}{k} = \binom{x-a}{k-1}$. Hence P is integer at every integer.

Hint 3. The same argument with $a = 0$ shows that every integer-valued polynomial is an integer combination of the $\binom{x}{k}$. The $\binom{x}{k}$ are integer-valued and have distinct degrees, so they are linearly independent. Therefore they form a basis of the $\mathbb{Z}$-module of integer-valued polynomials.

Problem 3

Hint 1. Cauchy–Schwarz: the case of two terms, $(ab + cd)^2 \leqslant (a^2 + c^2)(b^2 + d^2)$, is equivalent to $(ad - bc)^2 \geqslant 0$. For the step, put $A = \left(\sum_{i \leqslant n} a_i^2\right)^{1/2}$ and $B = \left(\sum_{i \leqslant n} b_i^2\right)^{1/2}$. Then $\sum_{i \leqslant n+1} a_i b_i \leqslant AB + a_{n+1} b_{n+1}$ by the induction hypothesis. Apply the two-term case to (A, a_{n+1}) and (B, b_{n+1}).

Hint 2. AM–GM forward: the case $n = 2$ is $(\sqrt{x} - \sqrt{y})^2 \geqslant 0$. From n to $2n$, apply the case n to each half and then the case 2 to the two means. This gives all powers of 2.

Hint 3. Backward step $n \to n-1$: given $x_1, \ldots, x_{n-1}$ with mean m, apply the n-term inequality to $x_1, \ldots, x_{n-1}, m$. This gives $m^n \geqslant x_1 \cdots x_{n-1}\, m$, hence $m^{n-1} \geqslant x_1 \cdots x_{n-1}$.

Hint 4. Why the scheme: a direct step $n \to n+1$ is awkward, because the mean of $n+1$ numbers does not split into the mean of n of them plus one term in a way that AM–GM for n handles directly. Doubling and adding the mean as an extra variable are the natural moves. Together they reach every n, since every n lies below some power of 2.

Problem 4

Hint 1. Class equation: $|G| = |Z(G)| + \sum_i [G : C_G(x_i)]$, where the sum runs over representatives of the conjugacy classes with more than one element. Each such index divides p^k and is greater than 1, so it is divisible by p.

Hint 2. Hence $p \mid |Z(G)|$, and since $e \in Z(G)$, the centre is non-trivial. An induction on k is not really needed here. If you want one, use it for the corollary that a p-group has normal subgroups of every order: pass to G/Z, of order $p^{k'}$ with $k' < k$.

Hint 3. Order p^2: if $Z \neq G$, then $|Z| = p$ and G/Z has order p, so it is cyclic. If G/Z is cyclic, generated by gZ, then every element is $g^i z$ with $z \in Z$, and any two such elements commute. So G is abelian, which contradicts $Z \neq G$.

Problem 5

($x^2 + y^2 + z^2 = 2xyz$)

Hint 1. Strengthen: show that $x^2 + y^2 + z^2 = 2^m xyz$ has no positive solutions for any $m \geqslant 1$. Look at the equation modulo 4, recalling that squares are 0 or 1 (mod 4).

Hint 2. The right side is even, so an even number of x, y, z are odd. If exactly two are odd, the left side is $\equiv 2 \pmod 4$, while the right side is divisible by 4 (it has a factor 2 and one even variable). So all three are even.

Hint 3. Substituting $x = 2x'$, $y = 2y'$, $z = 2z'$ gives $x'^2 + y'^2 + z'^2 = 2^{m+1} x'y'z'$, a solution of the same type with a smaller $x + y + z$. This is an infinite descent, so no solution exists.

($x^3 + 2y^3 = 4z^3$)

Hint 1. Assume a non-trivial integer solution with $|x|+|y|+|z|$ minimal (the hint covers integers, not just positive ones). Since x^3 is even, x is even.

Hint 2. With $x = 2x_1$: $8x_1^3 + 2y^3 = 4z^3$, so $4x_1^3 + y^3 = 2z^3$, and y is even. With $y = 2y_1$: $2x_1^3 + 4y_1^3 = z^3$, and z is even. With $z = 2z_1$: $x_1^3 + 2y_1^3 = 4z_1^3$, a smaller solution.

Problem 6

Hint 1. Equivalence $A \sim PAQ$ with P, Q invertible means doing row and column operations. If $A \neq 0$, move a non-zero entry to position $(1,1)$, scale it to 1, and clear the rest of the first row and column.

Hint 2. This gives $\begin{pmatrix} 1 & 0 \\ 0 & A' \end{pmatrix}$ with $\operatorname{rank} A' = r - 1$. Apply the induction hypothesis on the rank to A'.

Hint 3. Nilpotent N ($N^k = 0$): N is not invertible, so there is a $v \neq 0$ with $Nv = 0$. In a basis starting with v, $N = \begin{pmatrix} 0 & * \\ 0 & N' \end{pmatrix}$, and N' is nilpotent because the block-triangular form gives $(N')^k$ as the lower-right block of $N^k = 0$. Make N' strictly upper triangular by induction on the size. Conjugating by the matrix $\operatorname{diag}(1, S)$ keeps the first column zero.

Problem 7

Hint 1. On $[0,1] \cap \mathbb{Q}$ take $f(x) = 1/(x - 1/\sqrt{2})$: it is continuous and unbounded (as is $1/x$ on $(0,1]$). Any finitely many points of $[0,1]$ can be moved to rational points without changing their order, and an argument that looks only at finitely many points at a time cannot tell the two sets apart. So any proof must use completeness of $[0,1]$, which an induction over finitely many steps never sees.

Hint 2. Compactness proof: if f is unbounded, choose x_n with $|f(x_n)| > n$. By Bolzano–Weierstrass, a subsequence converges to some $x \in [0,1]$, and continuity at x gives a contradiction. Alternatively, let $s = \sup\{t : f \text{ is bounded on } [0,t]\}$. Show $s > 0$, that f is bounded on $[0,s]$, and that $s = 1$ ("real induction").

Hint 3. The rule: induction on $\mathbb{N}$ is available when the objects carry a natural-number parameter, and each object can be reduced to finitely many objects with smaller parameter. Behind this is well-foundedness. Over the reals, where descending sequences need not stop, it is replaced by sup, compactness, or a limit argument.

Problem 8

Hint 1. First prove Euler's formula $V - E + F = 2$ for connected plane graphs by induction on the number of edges. Removing an edge on a cycle merges two faces. If there is no cycle, the graph is a tree. Count edge–face incidences to get $E \leqslant 3V - 6$ for $V \geqslant 3$.

Hint 2. The sum of degrees is $2E \leqslant 6V - 12 < 6V$, so some vertex has degree at most 5. This property passes to subgraphs, which is what the induction on the number of vertices uses.

Hint 3. Five colours: remove a vertex v of degree $\leqslant 5$ and colour the rest by induction. If the neighbours use at most 4 colours, colour v. Otherwise v has five neighbours $v_1, \dots, v_5$ in cyclic order, with five different colours.

Hint 4. Kempe chains: consider the subgraph spanned by colours 1 and 3. If v_1 and v_3 lie in different components of it, swap the two colours in the component of v_1, which frees colour 1 for v. Otherwise a 1–3 path joins v_1 to v_3. Together with v it forms a closed curve that separates v_2 from v_4, so the 2–4 swap works instead.

Problem 9

Hint 1. Maximal ideal: apply Zorn's lemma to the proper ideals, ordered by inclusion (the zero ideal shows the family is non-empty). The union of a chain of ideals is an ideal, and it is proper because it does not contain 1. This is exactly where the unit is used.

Hint 2. Maximal independent set: the same argument, as in the class work. The union of a chain of independent sets is independent, since linear dependence involves only finitely many vectors.

Hint 3. Countably generated $V = \operatorname{span}\{v_1, v_2, \dots\}$: go through $v_1, v_2, \dots$ and keep v_n if it is not in the span of the vectors kept so far. By induction the kept vectors are independent and span $\operatorname{span}\{v_1, \dots, v_n\}$ at stage n. So they form a basis, and no choice principle beyond ordinary induction is needed.

Problem 10

Hint 1. Sperner in dimension n: take a triangulation of the simplex $[e_0, \dots, e_n]$. Each vertex gets a label i from the coordinates that are positive at that vertex (i.e. from the indices of the smallest face containing it). Claim: the number of fully labelled small simplices is odd. Dimension 1: along the segment, the label changes from 0 to 1 an odd number of times.

Hint 2. Induction step (door counting): call an $(n-1)$-face labelled $0, \dots, n-1$ a door. Count pairs (small simplex, door on its boundary). A fully labelled simplex has exactly one door, any other simplex has 0 or 2, and an interior door is counted twice.

Hint 3. The doors on the boundary all lie on the face $[e_0, \dots, e_{n-1}]$, and by the induction hypothesis there are an odd number of them. Hence the number of fully labelled simplices is odd, in particular non-zero.

Hint 4. Brouwer: for continuous $f : \Delta \to \Delta$, label a vertex v by some i with $v_i > 0$ and $f(v)_i \leqslant v_i$. Such an i exists, since the coordinates of both v and $f(v)$ sum to 1. The labelling is a Sperner labelling. Take finer and finer triangulations, and use compactness to get a limit point x of fully labelled simplices. Then $f(x)_i \leqslant x_i$ for all i, so $f(x) = x$. For $n = 1$ this is the intermediate value theorem applied to $f(x) - x$.